10TH CLASS BIOLOGY — NEW SYLLABUS (PECTAA)

Chapter 9 — Diseases and Immunity

Solved Exercise | New Book (PECTAA)

Exercise (Solved)

A. Select the correct answers for the following questions.

1. A doctor finds bacteria in a patient's blood sample. What type of disease is the patient suffering from?

(a) Infectious disease (b) Non-infectious disease
(c) Genetic disorder (d) Nutritional deficiency

Answer: (a) — Infectious diseases result from harmful organisms (pathogens) such as bacteria entering the body. Since bacteria are present in the blood, the patient is suffering from an infectious disease, which can spread from person to person.

2. Which of the following is an example of an infectious disease?

(a) Diabetes (b) Hypertension
(c) Influenza (d) Osteoporosis

Answer: (c) — Influenza (flu) is caused by the influenza virus, so it is an infectious disease. Diabetes and hypertension develop due to heredity and unhealthy lifestyle, and osteoporosis is a non-infectious disease of the bones.

3. A person develops rabies after being bitten by a dog. This is an example of:

(a) Non-infectious disease (b) Food-borne illness
(c) Zoonotic disease (d) Genetic disorder

Answer: (c) — A zoonotic disease is an infectious disease which is naturally transmitted from animals to humans. Rabies passes from the infected dog to the person through its bite, so it is a zoonotic disease.

4. Which of the following is a vector-borne disease?

(a) AIDS (b) Malaria
(c) Tuberculosis (d) Diabetes

Answer: (b) — Malaria is caused by a parasitic protozoan (Plasmodium) which is transmitted from one human to another by the female Anopheles mosquito. The mosquito is the vector. AIDS spreads by sexual contact and blood, and TB spreads through airborne droplets.

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5. A person develops dengue fever after a mosquito bite. In this case, the mosquito acts as:

(a) Host (b) Pathogen
(c) Vector (d) Reservoir

Answer: (c) — Vectors are living organisms which carry pathogens from one host to another. Dengue virus is carried by the Aedes mosquito, so the mosquito is the vector. The pathogen is the dengue virus and the host is the human being.

6. Which of the following is the first line of defence against pathogens?

(a) Antibodies (b) Skin and mucous membranes
(c) T cells (d) B cells

Answer: (b) — The first line of defence is the non-specific defence which includes the skin and the mucous membranes. Antibodies, T cells and B cells belong to the third line of defence, that is, acquired immunity.

7. Skin is categorized as:

(a) Non-specific defence (b) Specific defence
(c) Acquired immunity (d) Passive immunity

Answer: (a) — The skin is a non-specific defence, because it stops any pathogen regardless of the pathogen's identity. It is a physical barrier, and the oil and sweat glands in it inhibit the growth of many microbes. It is present by birth (innate immunity).

8. What triggers an allergic reaction?

(a) Pathogens (b) Allergens
(c) Antigens (d) Antibodies

Answer: (b) — An allergy is an abnormal immunological response of the body against a foreign substance which is normally not harmful. Such substances which cause allergies are called allergens, for example pollen, dust mites, certain foods, insect stings and some medicines.

B. Write short answers.

1. Define infectious diseases.

Infectious diseases are the communicable diseases which can spread from person to person, or from the environment to a person. They result from harmful organisms called pathogens entering the body; the common pathogens are viruses, bacteria, fungi, parasitic protozoans and worms. They spread through air, contaminated food or water, person-to-person contact and the bites of animals. Influenza, malaria, tuberculosis and COVID-19 are examples of infectious diseases.

2. Define zoonotic diseases. Also give examples.

A zoonotic disease is an infectious disease which is naturally transmitted from animals to humans. The zoonotic pathogens may be bacteria, viruses or other parasites, and humans get them through direct contact with animals, or through food, water or the environment. **Examples:** anthrax, which is caused by the bacterium Bacillus anthracis and spreads through contaminated hides and wool; bird flu, caused by influenza A virus and spread by close contact with infected birds; and Ebola, caused by Ebola virus and spread through the body fluids of infected humans or animals.

3. Enlist allergies with some common types.

An allergy is an abnormal immunological response of the body against a foreign substance which is normally not harmful; such substances are called allergens. The common allergens are pollen, dust mites, certain foods, insect stings and certain medicines. The major types of allergies are:

(i) Respiratory allergies: allergic rhinitis (hay fever), which causes sneezing, itching, nasal congestion and a runny nose, and asthma, which causes wheezing, shortness of breath and coughing.

(ii) Skin allergies: eczema (itchy, inflamed patches of skin) and contact dermatitis (redness, itching and blisters after touching nickel, latex or certain plants).

(iii) Food allergies: reactions from mild itching and swelling to difficult breathing and loss of consciousness, caused by peanuts, tree nuts, shellfish, fish, milk, eggs, soy and wheat.

(iv) Insect sting allergies: bee, wasp and ant stings cause local pain, swelling, hives and, in severe cases, anaphylaxis.

(v) Drug allergies: rashes and itching to severe anaphylactic reactions, caused by certain antibiotics, aspirin and NSAIDs.

4. Differentiate between innate and acquired immunity.

Innate immunity	Acquired immunity
It is present in the body by birth.	It is acquired during the lifetime of a person.
It includes the first and the second line of defence.	It includes the third line of defence.
It is a non-specific defence, so it works against any pathogen.	It is a specific defence, so it works against a particular pathogen.
Its examples are the skin, mucous membranes, phagocytes and inflammation.	Its examples are the antibodies made by B-cells and the action of T-cells; it may be active (vaccination) or passive (antibodies given to the patient).

5. Describe the role of skin as a component of defence system.

The skin is the main part of the first line of defence, and it works as a physical barrier which stops pathogens from entering the body. As long as the skin is unbroken, the bacteria and viruses cannot pass through it. In addition, the oil and sweat glands present in the skin produce secretions which inhibit the growth of many microbes, so the microbes are killed or their growth is stopped on the surface of the skin. In this way the skin gives a non-specific defence against all kinds of pathogens, regardless of their identity.

6. State the role of mucous membranes in defence.

Mucous membranes are epithelial tissues which protect the interior surfaces of the body, for example the respiratory and digestive systems and the urethra. These membranes secrete a sticky substance called mucus, which traps the pathogens entering through these openings. The trapped pathogens are then removed from the body, for example with the mucus of the nose and throat, so they cannot reach the inner tissues. Thus the mucous membranes act as a non-specific first line of defence.

C. Write answers in detail.

1. Describe infectious and non-infectious diseases and their types with examples.

Infectious diseases: these are the communicable diseases which can spread from person to person or from the environment to person. They result from pathogens entering the body, such as viruses, bacteria, fungi, parasitic protozoans and worms, and they spread through air, contaminated food and water, person-to-person contact and the bites of animals. Their types with examples are:

(i) Viral infections: hepatitis, flu, dengue, measles, HIV and COVID-19.

(ii) Bacterial infections: typhoid, strep throat, cholera and tuberculosis.

(iii) Fungal infections: candidiasis (yeast infection) and athlete's foot.

(iv) Infections by parasitic protozoans: malaria and amoebiasis.

(v) Infections by parasitic worms: roundworm and tapeworm infections.

Non-infectious diseases: these are the non-communicable diseases which do not transmit from one person to another. They result from heredity, unhealthy lifestyle, environmental factors and aging. Their types with examples are:

(i) Lifestyle diseases: heart diseases, hypertension, diabetes, cancer and stroke; mental health diseases like depression are also included.

(ii) Genetic diseases: these are caused by abnormalities in genes or chromosomes, for example haemophilia, thalassaemia and muscular dystrophy.

(iii) Diseases of aging: these result from the gradual decline of body organs or tissues, for example Alzheimer's disease, Parkinson's disease and arthritis.

2. Describe vector borne diseases with examples.

Vector-borne diseases are the infections which are transmitted by vectors, that is, living organisms which carry pathogens from one host to another. These diseases can be caused by bacteria, viruses or other parasites. The main examples are:

(i) Malaria: it is caused by a parasitic protozoan (Plasmodium). It primarily affects the liver and then the red blood cells, and if untreated it can lead to organ damage. It spreads from one human to another through the female Anopheles mosquito.

(ii) Dengue fever: it is caused by dengue virus, which is transmitted through the Aedes mosquito. It affects multiple organs, including the skin, blood vessels and lymph nodes.

(iii) Lyme disease: it is caused by the bacterium Borrelia burgdorferi, which spreads through ticks. The infection affects the skin, joints and nervous system.

(iv) Yellow fever: it is caused by yellow fever virus, which spreads through Aedes or Haemagogus mosquitoes. The infection first affects the liver and then involves the kidneys, heart and nervous system.

These diseases can be controlled by destroying the vectors, by using mosquito nets and repellents, and by keeping the surroundings clean so that the mosquitoes and ticks do not get places to breed.

3. Describe the roles of T-cells and B-cells in immunity.

B-cells: B-cells are a type of lymphocytes which provide the third line of defence (acquired immunity). When a pathogen enters the body, B-cells prepare special proteins called antibodies which are specific for that pathogen. The antibodies attach to the pathogen and help to kill it. This is called active immunity, and it may be natural, when the body makes the antibodies after an infection, or artificial, when the body is given a vaccine (vaccination). B-cells also keep the memory of the pathogen, so the body can fight it quickly if it enters again.

T-cells: T-cells are also lymphocytes, and they act by attacking the cells themselves. A special type of T-lymphocytes called natural killer cells kills the body cells which are infected by pathogens, and they also detect and kill cancer cells. In this way T-cells stop the pathogen from multiplying inside the body cells. Other T-cells also help the B-cells to make the right antibodies.

Thus B-cells work through antibodies present in the blood, while T-cells work directly on the infected or abnormal cells; both together give the body a strong specific defence. In passive immunity the ready-made antibodies are given to the person to combat specific microbes, but this immunity is short-lived.

4. Write a note on blood clotting.

Blood clotting, or coagulation, is the process which prevents excessive bleeding when a blood vessel is injured. Its main steps are the following.

(i) When a blood vessel is injured, the exposed collagen fibres activate the blood platelets. The activated platelets adhere to the collagen and to the damaged walls of the blood vessel, and in this way a temporary “platelet plug” is formed.

(ii) The platelets release special proteins called clotting factors, which convert prothrombin (a plasma protein) into thrombin.

(iii) Thrombin converts fibrinogen, another plasma protein, into fibrin strands. The fibrin strands weave through the platelet plug and form a blood clot.

(iv) The platelets within the clot contract and reduce the size of the wound. Once the vessel is repaired, a plasma protein breaks down the fibrin clot, which dissolves

the clot and restores the normal blood flow.

Prothrombin → Thrombin → Fibrin strands → Blood clot

In this way clotting stops the loss of blood, prevents the entry of pathogens through the wound and helps in the repair of the injured vessel.

D. Inquisitive questions

1. Why do infectious diseases spread faster in crowded places?

In crowded places the people are very close to one another, so the pathogens find it very easy to pass from one person to another. Most infectious diseases spread through airborne droplets which come out with coughing, sneezing and even talking, and in a crowd these droplets reach many people at once. The air in crowded and closed rooms is shared by everyone, and it is often not fresh, so the viruses and bacteria stay in it for a long time. In such places the drinking water and food are also shared, and the same surfaces are touched by many people, so diseases like cholera and typhoid spread quickly. Moreover, in crowded places many people live together, so the vectors such as mosquitoes get more persons to bite. That is why diseases spread faster in crowded places, and this is why we should keep the rooms ventilated, cover the mouth while coughing and keep the surroundings clean.

2. How can genetics increase a person's risk of developing non-infectious diseases?

Many non-infectious diseases are not caused only by lifestyle; they also have a genetic basis. Every person receives genes from both parents, and some of these genes carry abnormalities. If a person inherits such a faulty gene, the protein which it makes is also faulty, and the body system which depends on that protein works poorly. For example, haemophilia, thalassaemia and muscular dystrophy are caused by abnormalities in genes or chromosomes and pass from parents to children. In the same way, the tendency of diabetes, hypertension, heart disease and some cancers to run in families shows that certain alleles increase the risk of these diseases. A person who has such alleles is more likely to develop the disease than a person who does not have them, and unhealthy lifestyle can make the disease appear earlier. So genetics increases the risk, but a healthy diet and regular exercise can delay the onset of such diseases.

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3. Why do some people develop allergies while others do not?

Allergy is an abnormal immunological response of the body against a foreign substance which is normally not harmful, and it depends mainly on the immune system of the person. In some people the immune system becomes over-sensitive to an allergen such as pollen, dust mites or a certain food, and it treats this harmless substance as a dangerous pathogen. When the allergen enters again, the immune system releases inflammatory chemicals such as histamine, and these chemicals cause the symptoms of allergy: sneezing, itching, redness, swelling or difficulty in breathing. The tendency of the immune system to react in this way is often inherited, so allergies often run in families. The other person who has a less sensitive immune system, or who has never come in contact with that allergen, does not show any reaction. In this way both the genes of the person and the exposure to the allergen decide whether an allergy will develop or not.

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